

5. Data and AI in Public Markets

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Road Map

Introduction

Theoretical Framework (Part I): Information and Adverse Selection

Why Do People Trade? And Should They?

Theoretical Framework (Part II): Glosten-Milgrom Model

The Value of Information

Conclusion

Information in Financial Markets

- 1815: Nathan Rothschild learns of the British victory at Waterloo via carrier pigeons, allowing him to purchase UK bonds before their prices increase
- 2026: Investors use large data and AI to forecast future price movements
 - ▶ Market data, payment data, mobile geolocation, satellite imagery, web traffic, social media, etc.
- Information technology has changed, but the economics remain: accurate and timely information is valuable for investment decisions

Market for Information

- Generalist data providers focus on traditional datasets, especially market data (e.g., LSEG Data & Analytics, S&P Global, Bloomberg)
- Alternative data vendors include specialists (e.g., Kayrros in satellite imagery) and aggregators (e.g., Eagle Alpha); platforms such as Datarade list hundreds of providers and APIs
- Large and profitable market

Market value of S&P \$150 bn (versus Goldman Sachs \$180 bn)

Two-thirds of London Stock Exchange Group's revenues come selling data

Today

- How is information used in financial markets?
- How is information incorporated into prices?
- Who gains and who loses in this process?
- What is the value of information?
- Can there be too much information?

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Theoretical Framework

- Consider the market for a security
- Different investors may have different estimates of its value

Because they have different data, models, and expertise

Q. How is the price determined?

⇒ We need to understand:

1. How each investor behaves
2. How the aggregation of individual behaviors determines the market equilibrium

Theoretical Framework

- Investors $i = 1, \dots, N$ (funds, individuals, etc.)
- E_i is investor i 's own estimate of the asset value
- Each investor i
 - buys shares if $E_i > P$
 - sells or short-sells shares if $E_i < P$
 - does not trade if $E_i = P$

Case 1: Symmetric Information

- Suppose all investors have the same estimate: $E_i = 30$ for all i

Q. What will be the equilibrium price?

- *Definition:* The market is in **equilibrium** when supply and demand are equalized
- The equilibrium price is a. Less than 30 b. 30 c. More than 30

Case 1: Symmetric Information

- Suppose all investors have the same estimate: $E_i = 30$ for all i

Q. What will be the equilibrium price?

- *Definition:* The market is in **equilibrium** when supply and demand are equalized
- The equilibrium price is 30

Case 1: Symmetric Information

- Suppose good news arrives (e.g., better-than-expected quarterly earnings)
 - All investors update their estimate to $E_i = 32$
 - Price goes to 32
- Q. According to our theoretical framework, how much trading takes place to move the price from 30 to 32?
- a. None
 - b. Some: some investors buy until the price reaches 32
 - c. A lot: many investors trade and the price varies widely before stabilizing at 32

Case 1: Symmetric Information

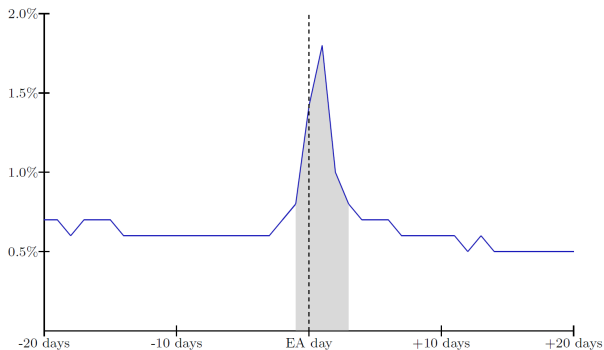
- Suppose good news arrives (e.g., better-than-expected quarterly earnings)
- All investors update their estimate to $E_i = 32$
- Price goes to 32

Q. According to our theoretical framework, how much trading takes place to move the price from 30 to 32?

- None! In our theoretical framework with symmetric information:
 1. No investors want to sell at 30 nor at any price below 32
 2. The price incorporates public information without any trading occurring

Evidence on Trading Upon Public Information

- Trading volume increases upon earnings announcements



Trading volume is computed as the percentage of shares that are turned over on each day, from 20 days preceding the announcement to 20 days after the announcement.

⇒ **Inconsistent with symmetric information.** We need to refine the theoretical framework

Case 2: Asymmetric Information

- Suppose different investors have different estimates because some investors are better informed than others
- $N = 2$ investors
 - ▶ Investor 1 has better information, more expertise, more resources (e.g., a hedge fund)
 - ▶ Investor 2 has less information (e.g., an individual)

Case 2: Asymmetric Information

- Initially, both investors have the same information and estimate the value of the security at 30
- Then, one of three events can happen, with equal probabilities:
 - New data reveal that the value is now 32. Investor 1 analyzes the data and learns the new value. Investor 2 is not aware of the new data
 - New data reveal that the value is now 28. Investor 1 analyzes the data and learns the new value. Investor 2 is not aware of the new data
 - The value remains 30. No new data

Q. What is the equilibrium of the market?

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Case 2: Asymmetric Information

- No trading occurs if investor 2 plays rationally

Investor 1 only buys when the price is too low and sells when the price is too high $\Rightarrow$ Investor 2 should never trade with investor 1

- No trading still prevails if:
 - ▶ $N \geq 2$ rational traders
 - ▶ Repeated trading rounds
 - ▶ Each trader has different information
 - ▶ Information is noisy

$\Rightarrow$ So-called “no-trade theorem”

- Another case of adverse selection (winner's curse)

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Trading Volume

- Evidence: Trading volume is large
 - ▶ Global stock market turnover (value of shares traded divided by market capitalization) $\approx 100\%$ per year



- ▶ Trading volume spikes upon news arrival (e.g., earnings announcements)

Explaining Trading

- Theory: The mere arrival of information does not logically lead to trading, whether the information is public (symmetric information) or private (asymmetric information)

Q. What is missing from the conceptual framework in the previous slides that could explain trading?

Explaining Trading

- Theory: The mere arrival of information does not logically lead to trading, whether the information is public (symmetric information) or private (asymmetric information)

Q. What is missing from the conceptual framework in the previous slides that could explain trading?

Trading motives not driven by (true) information

Two broad categories of such non-information trading motives

1. Less-Than-Rational Trading Motives

- Behavioral biases such as overconfidence, cognitive dissonance, self-attribution bias, limited attention, etc. (cf. class 1 with Daniel)
 - ▶ Explains high trading volume in general
 - ▶ Explains surge in trading around news events (more room for mistakes)
- Put differently: people may trade because they believe they are informed, but they aren't or they don't use information correctly
- Being non-rational $\neq$ having no information
 - ▶ One may have no information and correctly account for not having information (e.g., investor 2 in the trading game we played previously)
 - ▶ Conversely, one may have information but fail to use it properly (e.g., trading on information already incorporated into the price)

2. Rational Trading Motives: Liquidity and Hedging

- Liquidity motive

Receive a bonus and invest it $\Rightarrow$ buy stocks

Need cash to buy an apartment $\Rightarrow$ sell stocks

A company issues shares to finance capex $\Rightarrow$ sell

- Hedging/risk rebalancing motive

Take a job in the finance industry $\Rightarrow$ sell stocks of financial companies and buy stocks of tech companies to hedge

Standard investment advice is to reduce share of stocks in portfolio as one ages $\Rightarrow$ sell stocks and buy bonds over time¹

Companies buying or selling futures to hedge

- Those are legit economic motives, but they are not related to information about the value of securities
- These trading motives can explain only a small share of trading
 - ▶ Cannot explain why people buy and sell at daily frequencies
 - ▶ Cannot explain why trading surges around news events

¹This advice is debatable: blog post [Should you buy stocks when young or old?](#)

3. Informed Trading

- The third trading motive is information-driven trading (e.g., investor 1 in the trading game)
- Informed trading requires the presence of less-than-rational traders or liquidity traders/hedgers (otherwise the “no-trade theorem” applies)

Does Trading Create Value?

- We answered the question: Why do people trade?
- The next question is: Should they?
- Put differently: Does trading create value?
- The answer depends on the trading motives: less-than-rational trading decisions versus liquidity/hedging

Does Trading Create Value?

Case 1: Trading between informed investors and less-than-rational investors

- Informed investors win, irrational lose

Do retail investors make money? Hmm, no

<https://johanhombert.github.io/blog/20210205-day-trading>

⇒ Zero-sum game

Does Trading Create Value?

Case 2: Trading between informed traders and liquidity traders

- Example: An insurance company urgently needs €100 million to pay insurance claims, otherwise it would default on claims and incur penalties

A bank offers €100 million to buy corporate bonds worth €101 million from the insurer

⇒ Win-win: the insurer avoids default, the bank earns €1 million

- Conceptually: the banks provide (valuable) liquidity to the insurer

⇒ Value is created, not zero sum

Trading in Financial Markets — Summary

- Information alone does not generate trading
- Trading arises in the presence of:
 1. Less-than-rational traders
 - ▶ Profit opportunities for informed traders
 - ▶ Zero-sum game, no social value created
... at least not directly, but it may create social value through information production (more later)

OR

2. Liquidity and hedging needs
 - ▶ Profit opportunities for trading counterparties
 - ▶ Positive sum game, value is created

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Glosten-Milgrom Model

- A model that
 - ▶ encompasses different trader types and trading motives
 - ▶ shows how private information gets incorporated into prices
 - ▶ explains transaction costs
 - ▶ determines the private value of information

Glosten and Milgrom 1985, “Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously informed traders,” *Journal of Financial Economics*.

Modern presentation: Foucault, Roell and Pagano, *Market Liquidity: Theory, Evidence and Policy*, Chapter 3.

Organization of Trading

- Some securities trade in **organized exchanges** (Euronext, New-York Stock Exchange, Chicago Mercantile Exchange...)
- Trading on exchanges takes place in a **limit order book** using one of two types of trading order

Limit orders are offers to buy (or sell) at a given price

Market order are orders to buy (or sell) against a standing limit order

- **Market makers** (hedge funds, investment banks) are specialized in posting limit orders

Organization of Trading

- Other securities trade **over-the-counter** (OTC)
- In OTC markets, investors who want to trade contact **dealers** (Goldman Sachs, JP Morgan, BNP Paribas, etc.) to request a quote
- Dealers act as market makers. Quotes are similar to limit orders
- One key difference vs. organized exchange: limit orders in the book are public, dealers' quotes in OTC markets are not or less so

Glosten-Milgrom Model

- Initially, everyone agrees that the value of the security is 30
- A **market maker** posts **limit orders**
 - to buy at price 29.5 (the **bid** price)
 - to sell at price 30.5 (the **ask** price); the **bid-ask spread** is $30.5 - 29.5 = 1$
- Two types of investors may trade using **market orders**
 - Informed traders** have information about the security's value and are rational
 - Noise traders** have no information and trade for irrational reasons or liquidity/hedging reasons
- One of four events happens with equal probabilities
 - An informed trader finds out the fundamental value is 32 and buys
 - An informed trader finds out the fundamental value is 28 and sells
 - Fundamental value remains 30 and a noise trader buys
 - Fundamental value remains 30 and a noise trader sells
- Trading is anonymous: the market maker doesn't know if market orders come from an informed trader or a noise trader

Q. Suppose a trader buys. How much is the expected profit for the market maker?

a. Zero b. Minus 0.5 c. Plus 0.5

Glosten-Milgrom Model

- Market maker's profit when a trader buys = ask price minus true value
- Case 1: The buyer is an informed trader

$$\text{Market maker's profit} = 30.5 - 32 = -1.5$$

- Case 3: The buyer is a noise trader

$$\text{Market maker's profit} = 30.5 - 30 = +0.5$$

- Probability of case 1 conditional on a market buy order: 0.5
- Probability of case 3 conditional on a market buy order: 0.5

$$\Rightarrow \text{Market maker's expected profit} = 0.5 \times (-1.5) + 0.5 \times (+0.5) = -0.5$$

Glosten-Milgrom Model

- The market maker earns money against noise traders

... but loses money against informed traders (due to adverse selection: informed traders trade only when advantageous to them)

Q1. What ask price makes the market maker break even on average?

- a. 30 b. 31 c. 32

Glosten-Milgrom Model

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Q1. What ask price makes the market maker break even on average?

$$0.5 \times (\text{ask price} - 32) + 0.5 \times (\text{ask price} - 30) = 0$$

break-even ask price is 31

Q2. Can the market maker quote an ask price of 31.5 and earn an expected profit of $0.5 \times (31.5 - 32) + 0.5 \times (31.5 - 30) = 0.5$ per trade?

Glosten-Milgrom Model

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Q2. Can the market maker quote an ask price of 31.5 and earn an expected profit of $0.5 \times (31.5 - 32) + 0.5 \times (31.5 - 30) = 0.5$ per trade?

it depends on the intensity of competition between market makers

Competition Between Market Makers

- When competition between market makers is intense
 - ▶ If a market maker quotes an ask price at 31.5, another market maker will undercut at 31.4, then another will undercut at 31.3, etc., until the ask price is very close to 31 and market makers' profits are close to zero
- When competition between market makers is limited
 - ▶ Market makers are able to sustain an ask price significantly above 31 and earn significant profits
- In practice, competition between market makers is fairly intense in organized markets, less so in (opaque) OTC markets

Bid-Ask Spread

- Why is there a bid-ask spread?
 - Two reasons
1. **Adverse selection.** The bid-ask spread compensates market makers for the trading losses they incur when trading against informed traders
 - ▶ In previous example: break-even bid and ask are 29 and 31 $\Rightarrow$ bid-ask spread = 2 due to adverse selection
 2. **Market power.** If there is limited competition between market makers (such as in OTC markets), they can sustain a bid-ask spread larger than the break-even bid-ask spread

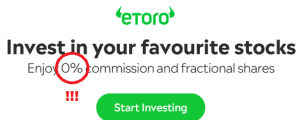
Transactions Costs and Liquidity

- The bid-ask spread creates **transaction costs** for investors
- Transaction costs grow more than proportionally with the number of shares traded because the price drifts against the trader. See practice problem 1
- The price change caused by a market order is called the **price impact** of the trade
- The **liquidity** of the market is the ease and cheapness of trading large quantities of shares
- High bid-ask spread $\Leftrightarrow$ less liquid market $\Leftrightarrow$ high price impact
- A market is less liquid when there is more asymmetric information and when market makers have more market power

Information and Trading

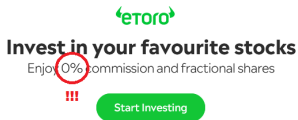
- The Glosten-Milgrom model provides a framework where (a) information generates trading and (b) trading incorporates information into the price
- Information arrival $\Rightarrow$ informed traders trade $\Rightarrow$ Market makers update the quoted price (bid and ask) because they know the order was informed with some probability
- This process is called **price discovery**

Zero-Commission Trading Apps



Q. What is the business model of zero-commission online brokers?

Zero-Commission Trading Apps



Q. What is the business model of zero-commission online brokers?

Payment for order flow

"If you don't pay, you are the product"

Payment For Order Flow

- It would be very valuable for market makers to know whether orders come from informed traders or noise traders
- Market makers pay retail brokerage firms to execute the trades of their clients

Citadel Securities, IMC paid US\$420m for retail flows as options surged



By **Etienne Mercuriali** November 11, 2025 8:55 Pm

Retail hunger for US options trading in Q3 has spiked further, with payment for order flow from brokers to market makers up by 40% compared with last year. Meanwhile, record [...]

See practice problem 3

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The Value of Information for Investors

Value of information for an investor = profit earned by trading on it. It depends on:

- Market liquidity

Liquid market $\Rightarrow$ low transaction costs $\Rightarrow$ higher profits

Liquidity depends on noise trading (see practice problem 2) and competition between market makers

- Uniqueness of information

If many investors have the same information, they all trade in the same direction at the same time $\Rightarrow$ market makers understand the trades are informed and adjust the price accordingly $\Rightarrow$ informed investors earn little

$\Rightarrow$ Implications for the value of data

1. Unlikely to make money using widely available information
2. Importance of unique data
3. Importance of speed: if you can trade before other informed investors, you can still exploit the information

Private Value vs. Social Value of Information

- The **private value** of information for the informed investor reflects trading profits, which comes at the expense of trading counterparties $\Rightarrow$ zero-sum
- The **social value** of information reflects overall value creation (not just for the informed investor)
- What is the social value of informed trading?
 - ▶ It makes security prices informative about fundamental values
 - ▶ Security prices are signals for companies' capital budgeting decisions
 - ▶ Therefore, more informative prices lead to better capital budgeting decisions

Security Prices and Capital Budgeting



Volume 32, Issue 7
July 2019

JOURNAL ARTICLE

Noisy Stock Prices and Corporate Investment



Get access >

Olivier Dessaint, Thierry Foucault, Laurent Frésard, Adrien Matray ✉

The Review of Financial Studies, Volume 32, Issue 7, July 2019, Pages
2625–2672, <https://doi.org/10.1093/rfs/hhy115>

article review here <https://johanhombert.github.io/blog/20210410-price-signal>

- Researchers estimate a multivariate regression model to explain capital expenditures decisions of US listed companies
- Main result: Capex increases when the stock price increases
 - ▶ Even if the price increases because of noise traders
 - ⇒ Informative prices are important for efficient capital budgeting
 - ▶ Capex of other companies in same industry also increases
 - ⇒ Informative price of one company matters for other companies

Private Value vs. Social Value of Information

- Private value of information $\neq$ social value of information
- $\Rightarrow$ Private incentives to produce information are generally not aligned with social efficiency
- If private value $>$ social value $\Rightarrow$ too much information produced from a social efficiency standpoint
- If private value $<$ social value $\Rightarrow$ too little information produced
- Which case prevails depends on the nature of information

Over-Investment in Short-Term Information?

- Information about the **short term** has lower social value
- Example: forecasting next day €/\$ exchange rate generates large profits, but is unlikely to matter for capital budgeting because the information will soon be public no matter what



With QuantCube's macroeconomic **nowcasting**, Discretionary Investors can create custom portfolios based on unique signals. Enhance macroeconomic research to drive Tactical Asset Allocation and create Smart Beta & Alpha Strategies effectively.

Many alternative data are used for **nowcasting**: estimate current economic variables that will be disclosed shortly (macro data, corporate earnings, etc.)

⇒ Excessive incentives to invest in nowcasting from a social efficiency perspective

Over-Investment in Short-Term Information?

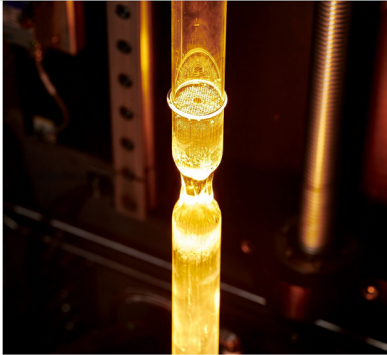
- Extreme example: high-frequency trading

THE WALL STREET JOURNAL.

MARKETS

**High-Frequency Traders
Push Closer to Light Speed
With Cutting-Edge Cables**

Firms aim to gain nanoseconds of advantage over rivals by using hollow-core fiber to convey data



A glass structure called an assembly is heated to ultrahigh temperatures, then stretched to form hollow-core fiber.

- \$300 million fiber-optic cable between Chicago and Wall Street to gain 3 milliseconds, quickly supplanted by microwave links, and so on. What is the social value of these investments?

Under-Investment in Long-Term Information?

- Information about the long term is valuable for capital budgeting (technologies, long-term risks, etc.)

... but more difficult on trade on it

1. Stock prices may take a long time to reflect the information
2. Startups developing new technologies are often not listed

⇒ Insufficient incentives to invest in long-term information?

Under-Investment in Long-Term Information?



Does Alternative Data Improve Financial Forecasting? The Horizon Effect

OLIVIER DESSAINT✉, THIERRY FOUCAULT, LAURENT FRESARD

First published: 07 March 2024

Over the past two decades, short-term forecasts have become more precise and long-term forecasts less precise in the last two decades

<https://johanhombert.github.io/blog/20210405-AI-informative-markets>

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Conclusion — How Financial Markets Create Value

- Primary markets: financing of real economy
 - ▶ Companies receive financing $\Rightarrow$ social value created
 - ▶ Households earn returns on their savings $\Rightarrow$ social value created
- Secondary markets: trading
 - ▶ For liquidity and hedging purposes $\Rightarrow$ social value created
 - ▶ For information purposes $\Rightarrow$ zero-sum game, but...

Conclusion — How Financial Markets Create Value

- Secondary markets: information production
 - ▶ Paradox: incentives to produce information come from the zero-sum exploitation of less-than-rational traders, but...
 - ▶ This process makes stock prices informative, which improves capital budgeting decisions ⇒ social value created
- ⇒ Private incentives to produce information may not align with social efficiency